CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Assignment 6: Serverless Architecture & Microservices Deployment

Course: Open Elective III [23OECS31]
Category: Assignments | Assignment #6 | Academic Year: 2026-27

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1. Overview & Objective:

Event-driven serverless computing, function invocation models, and API Gateway orchestration.

2. Assignment Problems & Instructions:

1. Analyze Function-as-a-Service FaaS event triggers, statelessness, and cold-start latency.
2. Compare monolithic application architectures with microservices communicating via asynchronous event buses.
3. Configure API Gateway rate limiting, throttling, and JWT token authorization mechanisms.
4. Compute cost analysis comparing an auto-scaled EC2 cluster against AWS Lambda for variable workloads.

Submission Guidelines & Verification:

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